

61. $\lim_{n \rightarrow \infty} \left(1 + \frac{1 + \frac{1}{2} + \dots + \frac{1}{n}}{n^2} \right)^n$ is equal to :

$\lim_{n \rightarrow \infty} \left(1 + \frac{1 + \frac{1}{2} + \dots + \frac{1}{n}}{n^2} \right)^n$ बराबर है :

- (A) $\frac{1}{e}$ (B) $\frac{1}{2}$ (C) 0 (D) 1

Ans. D

Sol. $L = e \lim_{n \rightarrow \infty} \left(\frac{1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}}{n} \right) = e^k$ (say)

Let $n = 2^p + \lambda$ $\lambda \in \{0, 1, 2, \dots, 2^p - 1\}$

$$1 + \left(\frac{1}{2} + \frac{1}{3} \right) + \left(\frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} \right) + \dots + \left(\frac{1}{2^{p-1}} + \frac{1}{2^{p-1}+1} + \dots + \frac{1}{2^p-1} \right) + \left(\frac{1}{2^p} + \frac{1}{2^p+1} + \dots + \frac{1}{2^p+\lambda} \right)$$

$$S < 1 + \left(\frac{1}{2} + \frac{1}{2} \right) + \left(\frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} \right) + \dots + \left(\frac{1}{2^p} + \frac{1}{2^p} + \dots + \frac{1}{2^p} \right)$$

(λ+1)time

$$S < 1 + 1 + 1 + 1 + \frac{\lambda + 1}{2^p} < p + 1$$

$$k \leq \lim_{p \rightarrow \infty} \frac{p+1}{2^p} = 0$$

$$S > \frac{1}{n} + \frac{1}{n} + \dots + \frac{1}{n} = 1$$

$$k \geq \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

So, $L = 1$

62. The real valued function $f(x) = \frac{\operatorname{cosec}^{-1}x}{\sqrt{x-[x]}}$, where $[x]$ denotes the greatest integer less than or equal to x , is

defined for all x belonging to :

- (A) All reals except integers
- (B) All non-integers except the interval $[-1, 1]$
- (C) All integers except $0, -1, 1$
- (D) All reals except the interval $[-1, 1]$

वास्तविक मान फलन $f(x) = \frac{\operatorname{cosec}^{-1}x}{\sqrt{x-[x]}}$, जहाँ $[x]$ महत्तम पूर्णांक $\leq x$ है, का प्रात है :

- (A) पूर्णाकों के अतिरिक्त सभी वास्तविक संख्याएँ
- (B) अंतराल $[-1, 1]$ के अतिरिक्त सभी वास्तविक संख्याएँ
- (C) $0, -1, 1$ के अतिरिक्त सभी पूर्णांक
- (D) अंतराल $[-1, 1]$ के अतिरिक्त सभी अपूर्णांक

Ans. B

Sol. $f(x) = \frac{\operatorname{cosec}^{-1}x}{\sqrt{x-[x]}}$

Domain of $\operatorname{cosec}^{-1}x$ is $x \in (-\infty, -1] \cup [1, \infty)$

for $\sqrt{x-[x]}$ or $\sqrt{\{x\}}$

$\{x\} \neq 0 \Rightarrow x \neq \text{Integer}$

So, final Domain is all non-integers except interval $[-1, 1]$

63. $\tan^{-1}\left(\frac{1+\sqrt{3}}{3+\sqrt{3}}\right) + \sec^{-1}\left(\sqrt{\frac{8+4\sqrt{3}}{6+3\sqrt{3}}}\right)$ is equal to :

$\tan^{-1}\left(\frac{1+\sqrt{3}}{3+\sqrt{3}}\right) + \sec^{-1}\left(\sqrt{\frac{8+4\sqrt{3}}{6+3\sqrt{3}}}\right)$ बराबर है :

- (A) $\frac{\pi}{6}$
- (B) $\frac{\pi}{2}$
- (C) $\frac{\pi}{4}$
- (D) $\frac{\pi}{3}$

Ans. D

Sol. $E = \tan^{-1}\left(\frac{1+\sqrt{3}}{\sqrt{3}(\sqrt{3}+1)}\right) + \sec^{-1}\left(\frac{16+8\sqrt{3}}{12+6\sqrt{3}}\right)^{1/2}$

$$\therefore \frac{16+8\sqrt{3}}{12+6\sqrt{3}} = \frac{4(\sqrt{3}+1)^2}{(3+\sqrt{3})^2}$$

$$\therefore \left(\frac{16+8\sqrt{3}}{12+6\sqrt{3}}\right)^{1/2} = \frac{2(\sqrt{3}+1)}{3+\sqrt{3}} = \frac{2}{\sqrt{3}}$$

$$E = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) + \sec^{-1}\left(\frac{2}{\sqrt{3}}\right) = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) + \cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$$

$$E = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) + \cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$$

$$= \frac{\pi}{6} + \frac{\pi}{6} = \frac{\pi}{3}$$

INTEGER TYPE QUESTIONS

64. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = \begin{cases} x+a, & x < 0 \\ |x-1|, & x \geq 0 \end{cases}$ and $g(x) = \begin{cases} x+1, & x < 0 \\ (x-1)^2 + b, & x \geq 0 \end{cases}$ where a, b are non-negative real numbers. If $(g \circ f)(x)$ is continuous for all $x \in \mathbb{R}$, then $a+b$ is equal to ____.

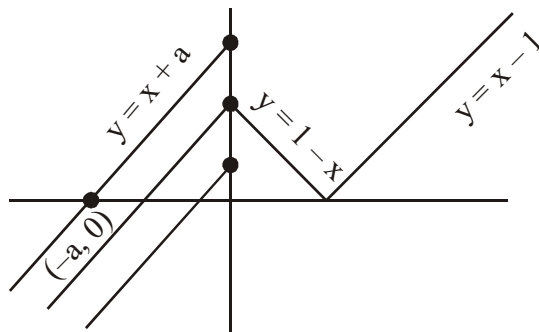
माना $f: \mathbb{R} \rightarrow \mathbb{R}$ तथा $g: \mathbb{R} \rightarrow \mathbb{R}$ $f(x) = \begin{cases} x+a, & x < 0 \\ |x-1|, & x \geq 0 \end{cases}$ तथा $g(x) = \begin{cases} x+1, & x < 0 \\ (x-1)^2 + b, & x \geq 0 \end{cases}$ द्वारा परिभाषित हैं, जहाँ a, b ऋणोत्तर वास्तविक संख्यायें हैं। यदि $(g \circ f)(x)$ सभी $x \in \mathbb{R}$, के लिए संतत है, तो $a+b$ बराबर ____।

Ans. 1

Sol. $f(x) = \begin{cases} x+a & x < 0 \\ |x-1| & x \geq 0 \end{cases}$

$$g(x) = \begin{cases} x+1 & x < 0 \\ (x-1)^2 + b & x \geq 0 \end{cases}$$

$$g(f(x)) = \begin{cases} f(x) + 1 & f(x) < 0 \\ (f(x) - 1)^2 + b & f(x) \geq 0 \end{cases}$$



$$g(f(x)) = \begin{cases} (x+a)+1 & x < -a \\ ((x+a)-1)^2 + b & -a \leq x < 0 \\ ((|x-1|)-1)^2 + b & 0 \leq x \end{cases}$$

For $g(f(x))$ to be continuous

$$g(f(-a^-)) = g(f(-a^+)) = g(f(-a))$$

$$\text{and } g(f(0^-)) = g(f(0^+)) = g(f(0))$$

$$1 = 1 + b \Rightarrow b = 0$$

$$(a-1)^2 + b = b \Rightarrow a = 1$$

$$a + b = 1$$

65. If the function $f(x) = \frac{\cos(\sin x) - \cos x}{x^4}$ is continuous at each point in its domain and $f(0) = \frac{1}{k}$, then k is :

यदि फलन $f(x) = \frac{\cos(\sin x) - \cos x}{x^4}$, अपने प्रांत के प्रत्येक बिन्दु पर संतत है $f(0) = \frac{1}{k}$ है, तो k बराबर है :

Ans. 6

Sol. $\frac{1}{k} = \lim_{x \rightarrow 0} \frac{2 \sin\left(\frac{\sin x + x}{2}\right) \sin\left(\frac{x - \sin x}{2}\right)}{x^4}$

$$= \lim_{x \rightarrow 0} \left(\frac{\sin x + x}{2} \right) \left(\lim_{x \rightarrow 0} \frac{x - \sin x}{2} \right)$$

$$= 1 \times \frac{1}{6} = \frac{1}{6}$$

$$k = 6$$

66. Let $f(x) = x \cdot \left[\frac{x}{2} \right]$, for $-10 < x < 10$, where $[t]$ denotes the greatest integer function. Then the number of points of discontinuity of f is equal to _____.

माना $f(x) = x \cdot \left[\frac{x}{2} \right]$, $-10 < x < 10$, है जहाँ $[t]$ महत्तम पूर्णांक फलन है, तो f के असंतत बिन्दुओं की संख्या है _____।

Ans. 8

Sol. Possible points of discontinuity are $(-8, -6, -4, -2, 0, 2, 4, 6, 8)$

checking $f(a^-) = f(a) = f(a^+)$, it is continuous at 0, discontinuous at rest.

Hence discontinuous at 8 points.

67. If $y = \sum_{k=1}^6 k \cos^{-1} \left\{ \frac{3}{5} \cos kx - \frac{4}{5} \sin kx \right\}$, then $\frac{dy}{dx}$ at $x = 0$ is _____.

यदि $y = \sum_{k=1}^6 k \cos^{-1} \left\{ \frac{3}{5} \cos kx - \frac{4}{5} \sin kx \right\}$, तो $x = 0$ पर $\frac{dy}{dx}$ का मान है _____.

Ans. 91

Sol. $y = \sum_{k=1}^6 k \cos^{-1} \left(\frac{3}{5} \cos kx - \frac{4}{5} \sin kx \right)$

$$\text{Let } \cos \theta = \frac{3}{5}$$

$$\sin \theta = \frac{4}{5}$$

$$y = \sum_{k=1}^6 k \cos^{-1} (\cos \theta \cdot \cos kx - \sin \theta \cdot \sin kx)$$

$$y = \sum_{k=1}^6 k \cos^{-1} (\cos(kx + \theta)) = \sum_{k=1}^6 k(kx + \theta)$$

$$\frac{dy}{dx} = \sum_{k=1}^6 k^2 = \frac{6(6+1)(12+1)}{6} = 91$$

68. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} x^5 \sin\left(\frac{1}{x}\right) + 5x^2, & x < 0 \\ 0, & x = 0 \\ x^5 \cos\left(\frac{1}{x}\right) + \lambda x^2, & x > 0 \end{cases}$$

The value of λ for which $f'(0)$ exists, is _____.

माना $f : \mathbb{R} \rightarrow \mathbb{R}$

$$f(x) = \begin{cases} x^5 \sin\left(\frac{1}{x}\right) + 5x^2, & x < 0 \\ 0, & x = 0 \\ x^5 \cos\left(\frac{1}{x}\right) + \lambda x^2, & x > 0 \end{cases}$$

द्वारा परिभाषित है। λ का मान जिसके लिए $f'(0)$ का अस्तित्व है, है _____।

Ans 5

Sol.

$$f(x) = \begin{cases} x^5 \sin\left(\frac{1}{x}\right) + 5x^2, & x < 0 \\ 0, & x = 0 \\ x^5 \cos\left(\frac{1}{x}\right) + \lambda x^2, & x > 0 \end{cases}$$

$$f(x) = \begin{cases} g(x) + 5x^2 & x < 0 \\ 0 & x = 0 \\ h(x) + \lambda x^2 & x > 0 \end{cases}$$

$$f'(x) = \begin{cases} g'(x) + 10x & x < 0 \\ 0 & x = 0 \\ h'(x) + 2\lambda x & x > 0 \end{cases}$$

$$g'(0) = h'(0) = 0$$

$$f''(x) = \begin{cases} g''(x) + 10 & x < 0 \\ 0 & x = 0 \\ h''(x) + 2\lambda & x > 0 \end{cases}$$

$$g''(0) = h''(0) = 0$$

$$f'(0) \text{ exists if } 10 = 2\lambda \Rightarrow \lambda = 5$$

69. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfy the equation $f(x+y) = f(x) \cdot f(y)$ for all $x, y \in \mathbb{R}$ and $f(x) \neq 0$ for any $x \in \mathbb{R}$. If the function f is differentiable at $x = 0$ and $f'(0) = 3$, then $\lim_{h \rightarrow 0} \frac{1}{h} (f(h) - 1)$ is equal to _____ .
- माना $f: \mathbb{R} \rightarrow \mathbb{R}$ समीकरण $f(x+y) = f(x) \cdot f(y) \forall x, y \in \mathbb{R}$ को संतुष्ट करता है तथा किसी भी $x \in \mathbb{R}$ के लिए $f(x) \neq 0$ है। यदि फलन f बिन्दु $x = 0$ पर अवकलनीय है तथा $f'(0) = 3$ है, तो $\lim_{h \rightarrow 0} \frac{1}{h} (f(h) - 1)$ बराबर है _____ ।

Ans. 3

Sol. $f(x+y) = f(x) \cdot f(y)$

$$f(x) = a^x$$

$$f'(x) = a^x \ln a$$

$$f'(0) = a^0 \times \ln a = 3$$

$$\Rightarrow a = e^3$$

$$f(x) = e^{3x}$$

$$\lim_{h \rightarrow 0} \frac{f(h) - 1}{h} = \lim_{h \rightarrow 0} \left(\frac{e^{3h} - 1}{3h} \right) \times 3 = 3$$

70. Let $S = \{1, 2, 3, 4, 5, 6\}$. Then the number of one-one function $f: S \rightarrow P(S)$, where $P(S)$ denote the power set of S , such that $f(n) \subset f(m)$ where $n < m$ is _____ .

माना $S = \{1, 2, 3, 4, 5, 6\}$ है तो ऐसे एकैकी फलनों $f: S \rightarrow P(S)$, जहाँ $P(S)$ समुच्चय S का घात समुच्चय $f(n) \subset f(m)$ है जब कि $n < m$ है, की संख्या है।

Ans. 3240

- Sol. Let $S = \{1, 2, 3, 4, 5, 6\}$, then the number of one-one functions, $f: S \rightarrow P(S)$, where $P(S)$ denotes the power set of S , such that $f(n) \subset f(m)$ where $n < m$ is

$$n(S) = 6$$

$$P(S) = \{\emptyset, \{1\}, \dots, \{6\}, \{1, 2\}, \dots, \{5, 6\}, \dots, \{1, 2, 3, 4, 5, 6\}\}$$

—64 elements

case – 1

$f(6) = S$ i.e. 1 option,

$f(5) =$ any 5 element subset A of S i.e. 6 options,

$f(4)$ = any 4 element subset B of A i.e. 5 options,
 $f(3)$ = any 3 element subset C of B i.e. 4 options,
 $f(2)$ = any 2 element subset D of C i.e. 3 options,
 $f(1)$ = any 1 element subset E of D or empty subset i.e. 3 options,

Total functions = 1080

Case – 2

$f(6)$ = any 5 element subset A of S i.e. 6 options,
 $f(5)$ = any 4 element subset B of A i.e. 5 options,
 $f(4)$ = any 3 element subset C of B i.e. 4 options,
 $f(3)$ = any 2 element subset D of C i.e. 3 options,
 $f(2)$ = any 1 element subset E of D i.e. 2 options,
 $f(1)$ = empty subset i.e. 1 option

Total functions = 720

Case – 3

$f(6) = S$

$f(5)$ = any 4 element subset A of 'S' i.e. 15 options,
 $f(4)$ = any 3 element subset B of A i.e. 4 options,
 $f(3)$ = any 2 element subset C of B i.e. 3 options,
 $f(2)$ = any 1 element subset D of C i.e. 2 options,
 $f(1)$ = empty subset i.e. 1 option

Total functions = 360

Case – 4

$f(6) = S$

$f(5)$ = any 5 element subset A of S i.e. 6 options,
 $f(4)$ = any 3 element subset B of A i.e. 10 options,
 $f(3)$ = any 2 element subset C of B i.e. 3 options,
 $f(2)$ = any 1 element subset D of C i.e. 2 options,
 $f(1)$ = empty subset i.e. 1 option

Total functions = 360

Case – 5

$f(6) = S$

$f(5)$ = any 5 element subset A of S i.e. 6 options,
 $f(4)$ = any 4 element subset B of A i.e. 5 options,
 $f(3)$ = any 2 element subset C of B i.e. 6 options,
 $f(2)$ = any 1 element subset D of C i.e. 2 options,
 $f(1)$ = empty subset i.e. 1 option

Total functions = 360

Case – 6

$$f(6) = S$$

$f(5)$ = any 5 element subset A of S i.e. 6 options,

$f(4)$ = any 4 element subset B of A i.e. 5 options,

$f(3)$ = any 3 element subset C of B i.e. 4 options,

$f(2)$ = any 1 element subset D of C i.e. 3 options,

$f(1)$ = empty subset i.e. 1 option

Total functions = 360

$\therefore$ Number of such functions = 3240

71. Let $S = \{1, 2, 3, 5, 7, 10, 11\}$. The number of non-empty subsets of S that have the sum of all elements a multiple of 3, is _____.

माना $S = \{1, 2, 3, 5, 7, 10, 11\}$ है। S के अतिरिक्त उपसमुच्चयों, जिनके सभी अवयवों का योग 3 का एक गुणज है, की संख्या है।

Ans. 43

Sol. Out of the given numbers one is $3k$ type and 3 of $3k + 1$ type and remaining three are $3k + 2$ type.

Number of subsets with 1 element = 1

1 of $3k$ type

Number of subsets with 2 elements

1 of $(3k + 1)$ type + 1 of $(3k + 2)$ type = 9

Number of subsets with 3 elements

1 of $3k$ type + 1 of $(3k + 1)$ type + 1 of $(3k + 2)$

type = 9

3 of $(3k + 1)$ type = 1

3 of $(3k + 2)$ type = 1

Number of subsets with 4 elements

1 of $3k$ type + 3 of $(3k + 1)$ type = 1

1 of $3k$ type + 3 of $(3k + 2)$ type = 1

2 of $(3k + 1)$ type + 2 of $(3k + 2)$ = 9

Number of subsets with 5 elements

1 of $3k$ type + 2 of $(3k + 1)$ type + 2 of $(3k + 2)$

type = 9

Number of subsets with 6 elements

3 of $3k + 1$ type + 3 of $3k + 2$ type = 1

The set itself = 1

Total = 43.

72. Let $A = \{a, b, c\}$ and $B = \{1, 2, 3, 4\}$. Then the number of elements in the set $C = \{f: A \rightarrow B \mid 2 \in f(A) \text{ and } f \text{ is not one-one}\}$ is _____.

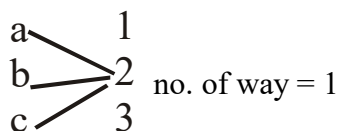
माना $A = \{a, b, c\}$ तथा $B = \{1, 2, 3, 4\}$ हैं, तो समुच्चय $C = \{f: A \rightarrow B \mid 2 \in f(A) \text{ तथा } f \text{ एकैकी नहीं है}\}$ के अवयवों की संख्या है _____.

Ans. 19

case : 1

when range of f contain only one element and $2 \in f(A)$

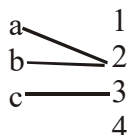
Then



Case : 2

when range of f contain exactly two elements and $2 \in f(A)$

for example



$$(1) {}^3C_1 \times \frac{3!}{2! \cdot 1!} \times 2! = 18$$

$$\text{Total} = 1 + 18 = 19$$

73. Let $A = \{-4, -3, -2, 0, 1, 3, 4\}$ and $R = \{(a, b) \in A \times A : b = |a| \text{ or } b^2 = a + 1\}$ be a relation on A . Then the minimum number of elements, that must be added to the relation R so that it becomes reflexive and symmetric, is _____.

माना $A = \{-4, -3, -2, 0, 1, 3, 4\}$ है तथा A पर एक संबंध $R = \{(a, b) \in A \times A : b = |a| \text{ या } b^2 = a + 1\}$ है। तो संबंध R में कम से कम कितने अवयव जोड़े जाएं, जिससे कि यह स्वतुल्य तथा सममित हो जाए _____

Ans. 7

Sol. $R = [(-4, 4), (-3, 3), (3, -2), (0, 1), (0, 0), (1, 1), (4, 4), (3, 3)]$

For reflexive, add $\Rightarrow (-2, -2), (-4, -4), (-3, -3)$

For symmetric, add $\Rightarrow (4, -4), (3, -3), (-2, 3), (1, 0)$

74. Let $f^1(x) = \frac{3x+2}{2x+3}$, $x \in \mathbb{R} - \left\{ \frac{-3}{2} \right\}$; For $n \geq 2$, define $f^n(x) = f^1 \circ f^{n-1}(x)$.

If $f^5(x) = \frac{ax+b}{bx+a}$, $\gcd(a,b) = 1$, then $a+b$ is equal to _____.

माना $f^1(x) = \frac{3x+2}{2x+3}$, $x \in \mathbb{R} - \left\{ \frac{-3}{2} \right\}$ है। $n \geq 2$, के लिए $f^n(x) = f^1 \circ f^{n-1}(x)$ द्वारा परिभाषित कीजिए। यदि $f^5(x) = \frac{ax+b}{bx+a}$, $\gcd(a,b) = 1$ है, तो $a+b$ बराबर है।

Ans. 3125

Sol. $f^1(x) = \frac{3x+2}{2x+3}$

$$\Rightarrow f^2(x) = \frac{13x+12}{12x+13}$$

$$\Rightarrow f^3(x) = \frac{63x+62}{62x+63}$$

$$\therefore f^5(x) = \frac{1563x+1562}{1562x+1563}$$

$$a+b = 3125$$

75. Let S be the set of points where the function, $f(x) = |2-|x-3||$, $x \in \mathbb{R}$, is not differentiable. Then

$$\sum_{x \in S} f(f(x))$$

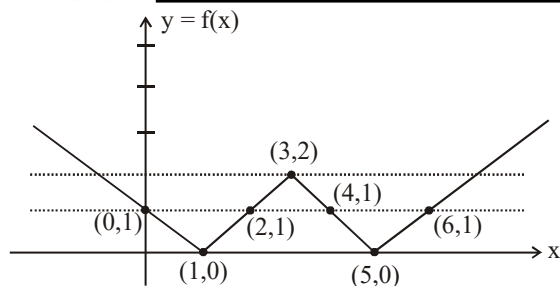
यदि S उन सभी बिन्दुओं का समुच्चय है, जिनके लिए फलन, $f(x) = |2-|x-3||$, $x \in \mathbb{R}$ अवकलनीय नहीं है, तो

$$\sum_{x \in S} f(f(x)) \text{ बराबर है } \text{-----} |$$

Ans. 3

Sol. $f(x) = |2-|x-3|| = ||x-3|-2|$

The graph of $f(x)$ is shown below :



76. If the sum of all the solution of $\tan^{-1}\left(\frac{2x}{1-x^2}\right) + \cot^{-1}\left(\frac{1-x^2}{2x}\right) = \frac{\pi}{3}$, $-1 < x < 1$, $x \neq 0$, is $\alpha - \frac{4}{\sqrt{3}}$, then α is equal to _____.

यदि $\tan^{-1}\left(\frac{2x}{1-x^2}\right) + \cot^{-1}\left(\frac{1-x^2}{2x}\right) = \frac{\pi}{3}$, $-1 < x < 1$, $x \neq 0$, के सभी हलों का योग $\alpha - \frac{4}{\sqrt{3}}$ है, तो α बराबर है।

Ans. 2

Sol. **Case I :** $x > 0$

$$\tan^{-1} \frac{2x}{1-x^2} + \tan^{-1} \frac{2x}{1-x^2} = \frac{\pi}{3}$$

$$x = 2 - \sqrt{3}$$

Case II : $x < 0$

$$\tan^{-1} \frac{2x}{1-x^2} + \tan^{-1} \frac{2x}{1-x^2} + \pi = \frac{\pi}{3}$$

$$x = \frac{-1}{\sqrt{3}} \Rightarrow \alpha = 2$$

77. Let $A = \{0, 3, 4, 6, 7, 8, 9, 10\}$ and R be the relation defined on A such that $R = \{(x, y) \in A \times A : x - y \text{ is odd positive integer or } x - y = 2\}$. The minimum number of elements that must be added to the relation R , so that it is a symmetric relation, is equal to _____.

माना $A = \{0, 3, 4, 6, 7, 8, 9, 10\}$ है तथा A पर एक संबंध R , $R = \{(x, y) \in A \times A : x - y \text{ विषम धनात्मक पूर्णांक है या } x - y = 2\}$. द्वारा परिभाषित है। संबंध R के सममित संबंध होने के लिए इसमें कम से कम कितने अवयव जोड़े जाएँ _____

Ans. 19

Sol. $A = \{10, 9, 8, 7, 6, 4, 3, 0\}$

$$R = \{(10, 9), (10, 8), (10, 7), (10, 3), (9, 8), (9, 7), (9, 6), (9, 4), (9, 0), (8, 7), (8, 6), (8, 3), (7, 6), (7, 4), (7, 0), (6, 4), (6, 3), (4, 3), (3, 0)\}$$

All the elements of R , (a, b) are of type $a > b$.

Hence we need to add total of 19 more elements to R to make in symmetric.

78. Let $A = \{1, 2, 3, 4\}$ and R be a relation on the set $A \times A$ defined by $R = \{(a, b), (c, d) : 2a + 3b = 4c + 5d\}$. Then the number of elements in R is _____.

यदि $A = \{1, 2, 3, 4\}$ तथा समुच्चय $A \times A$ में एक संबंध R इस प्रकार परिभाषित है कि $R = \{(a, b), (c, d) : 2a + 3b = 4c + 5d\}$ हो, तो R में अवयवों की संख्या होगी _____

Ans. 6

Sol. $A = \{1, 2, 3, 4\}$

$$R = \{(a, b), (c, d)\}$$

$$2a + 3b = 4c + 5d = \alpha \text{ let}$$

$$2a = \{2, 4, 6, 8\}$$

$$4c = \{4, 8, 12, 16\}$$

$$3b = \{3, 6, 9, 12\}$$

$$5d = \{5, 10, 15, 20\}$$

$$2a + 3b = \left\{ \begin{matrix} 5, 8, 11, 14 \\ 7, 10, 13, 16 \\ 9, 12, 15, 18 \\ 11, 14, 17, 20 \end{matrix} \right\} \quad 4c + 5d = \left\{ \begin{matrix} 9, 14, 19, 24 \\ 13, 18, \dots \\ 17, 22, \dots \\ 21, 26, \dots \end{matrix} \right\}$$

Possible value of $\alpha = 9, 13, 14, 17, 18$

Pairs of $\{(a, b), (c, d)\} = 6$

79. For some $a, b, c \in \mathbb{N}$, let $f(x) = ax - 3$ and $g(x) = x^b + c$, $x \in \mathbb{R}$. If $(f \circ g)^{-1}(x) = \left(\frac{x-7}{2}\right)^{\frac{1}{3}}$, then $(f \circ g)(ac) + (g \circ f)(b)$ is equal to _____.

किसी $a, b, c \in \mathbb{N}$ के लिए, माना $f(x) = ax - 3$ तथा $g(x) = x^b + c$, $x \in \mathbb{R}$ है। यदि $(f \circ g)^{-1}(x) = \left(\frac{x-7}{2}\right)^{\frac{1}{3}}$ है, तो

$(f \circ g)(ac) + (g \circ f)(b)$ बराबर है।

Ans. 2039

Sol. Let $f \circ g(x) = h(x)$

$$\Rightarrow h^{-1}(x) = \left(\frac{x-7}{2}\right)^{\frac{1}{3}}$$

$$\Rightarrow h(x) = f \circ g(x) = 2x^3 + 7$$

$$f \circ g(x) = a(xb + c) - 3$$

$$\Rightarrow a = 2, b = 3, c = 5$$

$$\Rightarrow f \circ g(ac) = f \circ g(10) = 2007$$

$$g(f(x)) = (2x - 3)^3 + 5$$

$$\Rightarrow g \circ f(b) = g \circ f(3) = 32$$

$$\Rightarrow \text{sum} = 2039$$

80. Let $A = \{1, 2, 3, 4, _, 10\}$ and $B = \{0, 1, 2, 3, 4\}$. The number of elements in the relation $R = \{(a, b) \in A \times A : 2(a - b)^2 + 3(a - b) \in B\}$ is _____.

माना $A = \{1, 2, 3, 4, _, 10\}$ और $B = \{0, 1, 2, 3, 4\}$ है। सम्बन्ध

$R = \{(a, b) \in A \times A : 2(a - b)^2 + 3(a - b) \in B\}$ में अवयवों की संख्या है _____

Ans. 18

Sol. $A = \{1, 2, 3, \dots, 10\}$

$$B = \{0, 1, 2, 3, 4\}$$

$$R = \{(a, b) \in A \times A : 2(a - b)^2 + 3(a - b) \in B\}$$

$$\text{Now } 2(a - b)^2 + 3(a - b) = (a - b)(2(a - b) + 3)$$

$$\Rightarrow a = b \text{ or } a - b = -2$$

When $a = b \Rightarrow 10$ order pairs

When $a - b = -2 \Rightarrow 8$ order pairs

$$\text{Total} = 18$$